

Satin Stitch Embroidery SOP

Fixed Setup: Isacord / Amann 40wt Polyester · 75/11 Needle · All Products
Version 1.0 — Compiled from sample testing (Alpine Flooring logo trials)

1. Fixed Constants (Do Not Change Per Job)

Item	Standard
Thread	Isacord (Amann) 40wt polyester, all needles
Needle size	75/11
Needle point (optional upgrade)	Keep 75/11 size, but stock SES light ballpoint 75/11 for polo & stretch fabrics — same size, gentler on knit loops. Sharp/R-point 75/11 stays for caps and wovens.
Bobbin	Standard 60–70wt bobbin thread

2. Size Bands (Decided by COLUMN WIDTH, not letter height)

Measure the narrowest satin column in the design. That band's settings apply to that object.

Band	Column Width	Typical Use
MICRO	< 1.5 mm	Small text 4.5–7 mm tall (e.g. KITCHEN & BATH)
SMALL	1.5 – 3 mm	Text 8–15 mm, thin borders
MEDIUM	3 – 6 mm	Large text, standard logo elements
WIDE	6 – 10 mm	Bold shapes, thick letters
VERY WIDE	> 10 mm	Split satin or convert to fill

40wt Hard Floor (applies to every product)

- Minimum letter height: **5 mm** (4.5 mm only with small-lettering fonts)
 - Minimum column width: **0.8 mm**; widen strokes to **≥ 1.1 mm** at digitizing stage whenever possible
 - Below the floor: enlarge text or simplify font — do not attempt. Add this check to art approval.
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3. PRODUCT SOP — STRUCTURED HAT / CAP

Substrate behavior: buckram-stiffened front, curved frame, fabric pushed outward while sewing. Highest distortion of the three products → highest compensation values.

Settings by Band — HAT

Parameter	MICRO <1.5mm	SMALL 1.5–3mm	MEDIUM 3–6mm	WIDE 6–10mm	VERY WIDE >10mm
Density (spacing, mm)	0.45 ✓ validated (open to 0.48 only if buildup appears)	0.42–0.45	0.38–0.42	0.37–0.40	0.37–0.40
Pull comp (absolute)	0.22 mm ✓ validated	0.18–0.20 mm	0.15–0.18 mm	0.18 mm	115–118%
Underlay	Center-run only, 1.5–1.8 mm length	Center-run + edge-run, offset 0.35	Edge-run + zigzag, offset 0.45	Edge-run + double zigzag, offset 0.5	Edge-run + tatami, offset 0.55
Short stitches	OFF	Off (or span ≥85)	On (90/75)	On (90/75)	On (90/75)
Auto-split	—	—	—	—	Split above 8 mm

✓ = confirmed on your own stitch-outs (white cap, KITCHEN & BATH sample).
These are the anchor values — change them only after a failed QC measurement,
not preemptively.

Cap-Specific Rules (all bands)

1. **Sequencing: center-out and bottom-up.** Digitize so the design sews from the middle of the cap outward toward the sides, and lower elements before upper. Never left-to-right across the whole front.
2. **Topping:** water-soluble topping (Solvay) over the sew area for MICRO and SMALL text — mandatory on unstructured/soft crowns, recommended on buckram.
3. **Backing:** buckram fronts usually need none extra; add one piece of tearaway on unstructured caps.
4. **Machine speed:** 600–650 spm for MICRO/SMALL lettering; ≤ 750 spm for the rest of the design.
5. **Hopping:** frame tight, crown seam centered; sew a basting/placement run if the design is close to the seam.
6. **Ties:** minimal/small tie setting on all lettering objects.

4. PRODUCT SOP — POLO (PIQUE KNIT)

Substrate behavior: textured, moderately stretchy knit. Stitches sink into the pique texture; moderate distortion. Mid-range compensation.

Settings by Band — POLO

Parameter	MICRO <1.5mm	SMALL 1.5–3mm	MEDIUM 3–6mm	WIDE 6–10mm	VERY WIDE >10mm
Density (spacing, mm)	0.45 ✓ validated (open to 0.48 only if buildup appears)	0.42–0.45	0.38–0.42	0.36–0.40	0.36–0.40
Pull comp (absolute)	0.22 mm ✓ validated	0.18 mm	0.15 mm	0.15–0.18 mm	115%
Underlay	Center-run only, 1.5–1.8 mm length	Center-run + edge-run, offset 0.35	Edge-run + zigzag, offset 0.45	Edge-run + double zigzag, offset 0.5	Edge-run + tatami, offset 0.55
Short stitches	OFF	Off (or span ≥85)	On (90/75)	On (90/75)	On (90/75)
Auto-split	—	—	—	—	Split above 8 mm

Polo-Specific Rules (all bands)

1. **Backing: 2.5–3 oz cutaway**, one piece, always. Never tearaway on pique — the design will distort after first wash.
 2. **Topping: mandatory for MICRO and SMALL** text on pique — this is what stops stitches sinking into the texture and reading as shrinkage. Recommended for MEDIUM.
 3. **Hooing:** hoop fabric + cutaway together, taut but NOT stretched. If the fabric snaps back when released, it was over-stretched and the design will pucker.
 4. **Needle point:** SES light ballpoint 75/11 preferred (prevents cut loops in the knit); sharp 75/11 acceptable.
 5. **Machine speed:** 650–700 spm for MICRO/SMALL lettering; normal speed elsewhere.
 6. **Ties:** minimal/small on all lettering.
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4B. PRODUCT SOP — BREATHABLE / PERFORMANCE POLO

(open-knit, moisture-wicking, mesh-texture polos)

Substrate behavior: open weave lets stitches sink between yarns (fuzzy edges), thin and shifty under the needle, and shows fabric through any coverage gap. Alignment settings carry over from standard polo — the extra work is all in coverage and stabilization.

Settings by Band — BREATHABLE POLO (differences from standard polo only)

Parameter	MICRO <1.5mm	SMALL 1.5–3mm	MEDIUM 3–6mm
Density — light thread on dark fabric	0.40–0.42	0.38–0.42	0.36–0.40
Density — tone-on-tone / dark on light	0.45 ✓	0.42–0.45	0.38–0.42
Pull comp	0.22 mm ✓ (unchanged)	0.18 mm	0.15 mm
Topping	MANDATORY — 1 layer Solvy; 2 layers on very open mesh	Mandatory	Mandatory
All other digitizing settings	Same as standard polo	Same	Same

Breathable-Polo-Specific Rules

1. **Topping is the primary crispness fix.** Re-test with topping BEFORE changing any digitizing values — sunken stitches look identical to bad digitizing but are a materials problem.
2. **Backing: no-show mesh cutaway** (standard cutaway shadows through thin breathable fabric). One layer normally; two crossed layers for dense designs.
3. **Hooing: zero stretch + anti-shift.** Hoop mesh taut, garment relaxed on top, secured with temporary spray adhesive or sticky-backed mesh. Loose knit shifting between penetrations = ragged edges.
4. **Needle: SES ballpoint 75/11** — matters most on this fabric class.
5. **Speed: 600 spm** for all lettering.
6. **Wash-test before production** — breathable knits show pucker only after first wash.

Color-Contrast Density Rule (applies to ALL products)

High contrast between thread and fabric (light-on-dark or dark-on-light where gaps expose fabric) → tighten density by **0.03–0.05 mm** from the validated value. Low contrast (tone-on-tone, white-on-white) → validated values as-is. Gaps that are invisible on matching fabric become specks on contrasting fabric at identical density.

5. PRODUCT SOP — STRETCHABLE / PERFORMANCE FABRIC

(polyester performance tees, lycra/spandex blends, athletic wear)

Substrate behavior: high stretch and recovery, thin, unstable. Highest risk of pucker and post-wash distortion. Needs maximum stabilization, lowest stitch stress.

Settings by Band — STRETCH

Parameter	MICRO <1.5mm	SMALL 1.5–3mm	MEDIUM 3–6mm	WIDE 6–10mm	VERY WIDE >10mm
Density (spacing, mm)	0.45–0.48 $\triangle$ start at 0.45, untested	0.45	0.40–0.42	0.38–0.40	Convert to fill — avoid wide satin
Pull comp (absolute)	0.22–0.25 mm $\triangle$ start at 0.22, untested	0.20 mm	0.18 mm	0.20 mm	—

Parameter	MICRO <1.5mm	SMALL 1.5–3mm	MEDIUM 3–6mm	WIDE 6–10mm	VERY WIDE >10mm
Underlay	Center-run only, 1.5–1.8 mm length	Center-run + edge-run, offset 0.35	Edge-run + zigzag, offset 0.45	Edge-run + double zigzag , offset 0.5	—
Short stitches	OFF	Off	On (90/75)	On (90/75)	—
Auto-split	—	—	—	Split above 7 mm	—

△ = not yet validated by a sample sew. No stretch-fabric trials were run during testing. Start from the hat/polo validated values (0.45 / 0.22), sew a sample, run the QC height measurement, and only increase comp toward 0.25 if curved letters measure short. Update this table with your confirmed values.

Stretch-Specific Rules (all bands)

1. **Backing: no-show mesh cutaway** (soft poly mesh), one piece; add a **second layer crosswise** for designs over ~8,000 stitches or dense coverage. Never tearaway.
 2. **Topping: mandatory for all text**, recommended for everything on very thin fabric.
 3. **Hooing: zero stretch**. Hoop the stabilizer taut and let the garment relax onto it (or use adhesive-backed mesh / spray adhesive). Any stretch in the hoop = guaranteed pucker after release.
 4. **Needle point: SES light ballpoint 75/11 strongly recommended** — sharp needles cut lycra fibers and cause runs/holes.
 5. **Design-side rule**: prefer smaller total stitch counts; large dense satin blocks will pucker regardless of settings. Push back on artwork with heavy coverage.
 6. **Machine speed**: 600–650 spm throughout, not just lettering.
 7. **Ties**: minimal/small everywhere.
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5B. WATER-SOLUBLE TOPPING RULES (Solvay / Avalon / Wash-Away Film)

Compiled from published guidance by Sulky, Madeira, and Gunold, plus commercial trade sources.

The Decision Test

Topping exists to hold stitches ON TOP of the fabric during sew-out. Ask one question: **"Can stitches sink into or get lost in this surface?"** → YES = use topping. NO = skip it.

USE Topping

Situation	Why
Napped / pile fabrics: terry, fleece, velour, faux fur, suede, corduroy	Classic case — pile swallows stitches
Knits, incl. piqué polos, golf shirts, breathable/performance knits	Manufacturer-named use case; industry standard is "always on piqué"
MICRO / SMALL band lettering — on ANY fabric, even smooth ones	Both Sulky and Madeira explicitly name small lettering & fine detail clarity
Small text stitched on top of fill areas	Text sinks into fill texture; topping over the fill gives clean letter edges
Stretchy activewear	Acts as a secondary stabilizer, washes away after
Customer-supplied locked file digitized for a different fabric	Topping rescues the sew-out without editing the DST

SKIP / DO NOT Use Topping

Situation	Rule
Garment cannot be washed / is water-sensitive	HARD PROHIBITION — check care label first. Dry-clean-only, some leathers,

Situation	Rule
	water-marking fabrics → use heat-away film instead
Colours that may run	Do not mist-spray or leave film wet on garment; never leave embroidery damp more than a few minutes
Smooth stable wovens (twill, canvas, denim, oxford) with MEDIUM+ elements	No pile to sink into — wasted material and removal labor. Exception: MICRO/SMALL lettering still benefits (see above)
Buckram cap fronts — logo & large text	Not needed. Use only for MICRO/SMALL lettering on caps
Borderline fabrics	A/B sample (with vs without) is the tiebreaker

Layers & Weight

- **One layer is the default.** Two layers only for very open mesh knits or problem/locked designs.
- Standard-weight film (Solvay / Avalon 20-micron class) for normal work; heavier film (Super Solvay class) only for very dense designs or extra-fluffy nap.

Removal SOP

1. Tear away large excess by hand close to the stitching after sew-out.
2. Small remnants: damp cloth laid on top + iron dry, or light mist on tiny areas. First customer wash removes the rest.
3. **Letter counters (O, A, P, B, R) trap topping** — budget removal time for these on small text. Volume trick: press a damp ball made of leftover topping scraps onto stuck bits; film lifts film.
4. Do not stack or pile garments while topping residue is still damp.

6. UNIVERSAL RULES — ALL PRODUCTS, ALL SIZES

Junctions (K, N, &, X, Y, W, M)

- Overlap/tuck-under at column intersections: **0.3–0.5 mm only**.
- Sewing order within a letter: diagonals/crossbars FIRST, verticals LAST on top.
- Taper/miter diagonal column ends parallel to the vertical edge — no blunt square cuts into a joint.

- Move tie-in/tie-out away from junction points (mid-column or outer corner).
- Optional: diagonal density 0.05 mm looser than verticals to reduce joint pile.

Micro-Text Checklist (every job containing <1.5 mm columns)

- ☐ Letter height ≥ 5 mm (art approval gate)
- ☐ Stroke width widened to ≥ 1.1 mm where font allows
- ☐ Small-lettering font used where possible
- ☐ Short stitches OFF on these objects
- ☐ Ties set to minimal
- ☐ Care label checked — garment is washable (topping is prohibited on non-washable items; use heat-away instead)
- ☐ Topping cut and ready
- ☐ Speed reduced per product rule
- ☐ Test sew on scrap of same fabric before production run

Quality Check After Sample

- Measure finished height of a vertical-spine letter (I, stem of K) vs a curved/horizontal letter (B, C, O).
- **Pass: difference ≤ 0.1 mm.** Fail → increase pull comp by +0.03 mm and resew.
- Check junctions of N, K, & under magnification for lumps.
- For polo/stretch: wash-test one sample before approving large orders.

7. Quick Reference — What Changes Between Products

Factor	HAT	POLO	STRETCH
Pull comp (MICRO band)	0.22 mm ✓ validated	0.22 mm ✓ validated	Start 0.22, escalate to 0.25 only if QC fails △
Backing	Buckram / tearaway	2.5–3 oz cutaway	No-show mesh cutaway ×1–2
Topping	Micro/Small text	Micro/Small mandatory, Medium recommended	All text mandatory
Hooping	Tight frame, center seam	Taut, never stretched	Stabilizer taut, garment relaxed

Factor	HAT	POLO	STRETCH
Sequencing	Center-out, bottom-up	Normal	Normal, low stitch count
Speed	600–650 small text	650–700 small text	600–650 everything
Needle point	Sharp 75/11	SES ballpoint 75/11 preferred	SES ballpoint 75/11 strongly recommended
Wide satin limit	Split > 8 mm	Split > 8 mm	Split > 7 mm; avoid very wide

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1. Fixed Constants (Do Not Change Per Job)

Item	Standard
Thread	Isacord (Amann) 40wt polyester, all needles
Needle size	75/11
Needle point (optional upgrade)	Keep 75/11 size, but stock SES light ballpoint 75/11 for polo & stretch fabrics — same size, gentler on knit loops. Sharp/R-point 75/11 stays for caps and wovens.
Bobbin	Standard 60–70wt bobbin thread

2. Size Bands (Decided by COLUMN WIDTH, not letter height)

Measure the narrowest satin column in the design. That band's settings apply to that object.

Band	Column Width	Typical Use
MICRO	< 1.5 mm	Small text 4.5–7 mm tall (e.g. KITCHEN & BATH)
SMALL	1.5 – 3 mm	Text 8–15 mm, thin borders
MEDIUM	3 – 6 mm	Large text, standard logo elements
WIDE	6 – 10 mm	Bold shapes, thick letters
VERY WIDE	> 10 mm	Split satin or convert to fill

40wt Hard Floor (applies to every product)

- Minimum letter height: **5 mm** (4.5 mm only with small-lettering fonts)
- Minimum column width: **0.8 mm**; widen strokes to **≥ 1.1 mm** at digitizing stage whenever possible
- Below the floor: enlarge text or simplify font — do not attempt. Add this check to art approval.

3. PRODUCT SOP — STRUCTURED HAT / CAP

Substrate behavior: buckram-stiffened front, curved frame, fabric pushed outward while sewing. Highest distortion of the three products → highest compensation values.

Settings by Band — HAT

Parameter	MICRO ≤1.5mm	SMALL 1.5–3mm	MEDIUM 3–6mm	WIDE 6–10mm	VERY WIDE ≥10mm
Density (spacing, mm)	0.45 ✓ validated (open to 0.48 only if buildup appears)	0.42–0.45	0.38–0.42	0.37–0.40	0.37–0.40

Parameter	MICRO <1.5mm	SMALL 1.5–3mm	MEDIUM 3–6mm	WIDE 6–10mm	VERY WIDE >10mm
Pull comp (absolute)	0.22 mm ✓ validated	0.18–0.20 mm	0.15–0.18 mm	0.18 mm	115–118%
Underlay	Center-run only, 1.5–1.8 mm length	Center-run + edge-run, offset 0.35	Edge-run + zigzag, offset 0.45	Edge-run + double zigzag, offset 0.5	Edge-run + tatami, offset 0.55
Short stitches	OFF	Off (or span ≥85)	On (90/75)	On (90/75)	On (90/75)
Auto-split	—	—	—	—	Split above 8 mm

✓ = confirmed on your own stitch-outs (white cap, KITCHEN & BATH sample).
These are the anchor values — change them only after a failed QC measurement,
not preemptively.

Cap-Specific Rules (all bands)

1. **Sequencing: center-out and bottom-up.** Digitize so the design sews from the middle of the cap outward toward the sides, and lower elements before upper. Never left-to-right across the whole front.
2. **Topping:** water-soluble topping (Solvay) over the sew area for MICRO and SMALL text — mandatory on unstructured/soft crowns, recommended on buckram.
3. **Backing:** buckram fronts usually need none extra; add one piece of tearaway on unstructured caps.
4. **Machine speed:** 600–650 spm for MICRO/SMALL lettering; ≤ 750 spm for the rest of the design.
5. **Hooping:** frame tight, crown seam centered; sew a basting/placement run if the design is close to the seam.
6. **Ties:** minimal/small tie setting on all lettering objects.

4. PRODUCT SOP — POLO (PIQUE KNIT)

Substrate behavior: textured, moderately stretchy knit. Stitches sink into the pique texture; moderate distortion. Mid-range compensation.

Settings by Band — POLO

Parameter	MICRO <1.5mm	SMALL 1.5–3mm	MEDIUM 3–6mm	WIDE 6–10mm	VERY WIDE >10mm
Density (spacing, mm)	0.45 ✓ validated (open to 0.48 only if buildup appears)	0.42–0.45	0.38–0.42	0.36–0.40	0.36–0.40
Pull comp (absolute)	0.22 mm ✓ validated	0.18 mm	0.15 mm	0.15–0.18 mm	115%
Underlay	Center-run only, 1.5–1.8 mm length	Center-run + edge-run, offset 0.35	Edge-run + zigzag, offset 0.45	Edge-run + double zigzag, offset 0.5	Edge-run + tatami, offset 0.55
Short stitches	OFF	Off (or span ≥85)	On (90/75)	On (90/75)	On (90/75)
Auto-split	—	—	—	—	Split above 8 mm

Polo-Specific Rules (all bands)

1. **Backing: 2.5–3 oz cutaway**, one piece, always. Never tearaway on pique — the design will distort after first wash.
2. **Topping: mandatory for MICRO and SMALL** text on pique — this is what stops stitches sinking into the texture and reading as shrinkage. Recommended for MEDIUM.
3. **Hooping:** hoop fabric + cutaway together, taut but NOT stretched. If the fabric snaps back when released, it was over-stretched and the design will pucker.
4. **Needle point:** SES light ballpoint 75/11 preferred (prevents cut loops in the knit); sharp 75/11 acceptable.
5. **Machine speed:** 650–700 spm for MICRO/SMALL lettering; normal speed elsewhere.
6. **Ties:** minimal/small on all lettering.

4B. PRODUCT SOP — BREATHABLE / PERFORMANCE POLO

(open-knit, moisture-wicking, mesh-texture polos)

Substrate behavior: open weave lets stitches sink between yarns (fuzzy edges), thin and shifty under the needle, and shows fabric through any coverage gap. Alignment settings carry over from standard polo — the extra work is all in coverage and stabilization.

Settings by Band — BREATHABLE POLO (differences from standard polo only)

Parameter	MICRO <1.5mm	SMALL 1.5–3mm	MEDIUM 3–6mm
Density — light thread on dark fabric	0.40–0.42	0.38–0.42	0.36–0.40
Density — tone-on-tone / dark on light	0.45 ✓	0.42–0.45	0.38–0.42
Pull comp	0.22 mm ✓ (unchanged)	0.18 mm	0.15 mm
Topping	MANDATORY — 1 layer Solvy; 2 layers on very open mesh	Mandatory	Mandatory
All other digitizing settings	Same as standard polo	Same	Same

Breathable-Polo-Specific Rules

1. **Topping is the primary crispness fix.** Re-test with topping BEFORE changing any digitizing values — sunken stitches look identical to bad digitizing but are a materials problem.
2. **Backing: no-show mesh cutaway** (standard cutaway shadows through thin breathable fabric). One layer normally; two crossed layers for dense designs.
3. **Hoopings: zero stretch + anti-shift.** Hoop mesh taut, garment relaxed on top, secured with temporary spray adhesive or sticky-backed mesh. Loose knit shifting between penetrations = ragged edges.
4. **Needle: SES ballpoint 75/11** — matters most on this fabric class.
5. **Speed: 600 spm** for all lettering.
6. **Wash-test before production** — breathable knits show pucker only after first wash.

Color-Contrast Density Rule (applies to ALL products)

High contrast between thread and fabric (light-on-dark or dark-on-light where gaps expose fabric) → tighten density by **0.03–0.05 mm** from the validated value. Low contrast

(tone-on-tone, white-on-white) → validated values as-is. Gaps that are invisible on matching fabric become specks on contrasting fabric at identical density.

5. PRODUCT SOP — STRETCHABLE / PERFORMANCE FABRIC

(polyester performance tees, lycra/spandex blends, athletic wear)

Substrate behavior: high stretch and recovery, thin, unstable. Highest risk of pucker and post-wash distortion. Needs maximum stabilization, lowest stitch stress.

Settings by Band — STRETCH

Parameter	MICRO <1.5mm	SMALL 1.5–3mm	MEDIUM 3–6mm	WIDE 6–10mm	VERY WIDE >10mm
Density (spacing, mm)	0.45–0.48 $\triangle$ start at 0.45, untested	0.45	0.40–0.42	0.38–0.40	Convert to fill — avoid wide satin
Pull comp (absolute)	0.22–0.25 mm $\triangle$ start at 0.22, untested	0.20 mm	0.18 mm	0.20 mm	—
Underlay	Center-run only, 1.5–1.8 mm length	Center-run + edge-run, offset 0.35	Edge-run + zigzag, offset 0.45	Edge-run + double zigzag , offset 0.5	—
Short stitches	OFF	Off	On (90/75)	On (90/75)	—
Auto-split	—	—	—	Split above 7 mm	—

$\triangle$ = not yet validated by a sample sew. No stretch-fabric trials were run during testing. Start from the hat/polo validated values (0.45 / 0.22), sew a sample, run the QC height measurement, and only increase comp toward 0.25 if curved letters measure short. Update this table with your confirmed values.

Stretch-Specific Rules (all bands)

1. **Backing: no-show mesh cutaway** (soft poly mesh), one piece; add a **second layer crosswise** for designs over ~8,000 stitches or dense coverage. Never tearaway.
 2. **Topping: mandatory for all text**, recommended for everything on very thin fabric.
 3. **Hooing: zero stretch**. Hoop the stabilizer taut and let the garment relax onto it (or use adhesive-backed mesh / spray adhesive). Any stretch in the hoop = guaranteed pucker after release.
 4. **Needle point: SES light ballpoint 75/11 strongly recommended** — sharp needles cut lycra fibers and cause runs/holes.
 5. **Design-side rule**: prefer smaller total stitch counts; large dense satin blocks will pucker regardless of settings. Push back on artwork with heavy coverage.
 6. **Machine speed**: 600–650 spm throughout, not just lettering.
 7. **Ties**: minimal/small everywhere.
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5B. WATER-SOLUBLE TOPPING RULES (Solvay / Avalon / Wash-Away Film)

Compiled from published guidance by Sulky, Madeira, and Gunold, plus commercial trade sources.

The Decision Test

Topping exists to hold stitches ON TOP of the fabric during sew-out. Ask one question: **"Can stitches sink into or get lost in this surface?"** → YES = use topping. NO = skip it.

USE Topping

Situation	Why
Napped / pile fabrics: terry, fleece, velour, faux fur, suede, corduroy	Classic case — pile swallows stitches
Knits, incl. piqué polos, golf shirts, breathable/performance knits	Manufacturer-named use case; industry standard is "always on piqué"
MICRO / SMALL band lettering — on ANY fabric, even smooth ones	Both Sulky and Madeira explicitly name small lettering & fine detail clarity
Small text stitched on top of fill areas	Text sinks into fill texture; topping over the fill gives clean letter edges

Situation	Why
Stretchy activewear	Acts as a secondary stabilizer, washes away after
Customer-supplied locked file digitized for a different fabric	Topping rescues the sew-out without editing the DST

SKIP / DO NOT Use Topping

Situation	Rule
Garment cannot be washed / is water-sensitive	HARD PROHIBITION — check care label first. Dry-clean-only, some leathers, water-marking fabrics → use heat-away film instead
Colours that may run	Do not mist-spray or leave film wet on garment; never leave embroidery damp more than a few minutes
Smooth stable wovens (twill, canvas, denim, oxford) with MEDIUM+ elements	No pile to sink into — wasted material and removal labor. Exception: MICRO/SMALL lettering still benefits (see above)
Buckram cap fronts — logo & large text	Not needed. Use only for MICRO/SMALL lettering on caps
Borderline fabrics	A/B sample (with vs without) is the tiebreaker

Layers & Weight

- **One layer is the default.** Two layers only for very open mesh knits or problem/locked designs.
- Standard-weight film (Solvay / Avalon 20-micron class) for normal work; heavier film (Super Solvy class) only for very dense designs or extra-fluffy nap.

Removal SOP

1. Tear away large excess by hand close to the stitching after sew-out.
2. Small remnants: damp cloth laid on top + iron dry, or light mist on tiny areas. First customer wash removes the rest.
3. **Letter counters (O, A, P, B, R) trap topping** — budget removal time for these on small text. Volume trick: press a damp ball made of leftover topping scraps onto stuck bits; film lifts film.

4. Do not stack or pile garments while topping residue is still damp.
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6. UNIVERSAL RULES — ALL PRODUCTS, ALL SIZES

Junctions (K, N, &, X, Y, W, M)

- Overlap/tuck-under at column intersections: **0.3–0.5 mm only**.
- Sewing order within a letter: diagonals/crossbars FIRST, verticals LAST on top.
- Taper/miter diagonal column ends parallel to the vertical edge — no blunt square cuts into a joint.
- Move tie-in/tie-out away from junction points (mid-column or outer corner).
- Optional: diagonal density 0.05 mm looser than verticals to reduce joint pile.

Micro-Text Checklist (every job containing <1.5 mm columns)

- ☐ Letter height ≥ 5 mm (art approval gate)
- ☐ Stroke width widened to ≥ 1.1 mm where font allows
- ☐ Small-lettering font used where possible
- ☐ Short stitches OFF on these objects
- ☐ Ties set to minimal
- ☐ Care label checked — garment is washable (topping is prohibited on non-washable items; use heat-away instead)
- ☐ Topping cut and ready
- ☐ Speed reduced per product rule
- ☐ Test sew on scrap of same fabric before production run

Quality Check After Sample

- Measure finished height of a vertical-spine letter (I, stem of K) vs a curved/horizontal letter (B, C, O).
 - **Pass: difference ≤ 0.1 mm.** Fail → increase pull comp by +0.03 mm and resew.
 - Check junctions of N, K, & under magnification for lumps.
 - For polo/stretch: wash-test one sample before approving large orders.
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7. Quick Reference — What Changes Between Products

Factor	HAT	POLO	STRETCH
Pull comp (MICRO band)	0.22 mm ✓ validated	0.22 mm ✓ validated	Start 0.22, escalate to 0.25 only if QC fails △
Backing	Buckram / tearaway	2.5–3 oz cutaway	No-show mesh cutaway ×1–2
Topping	Micro/Small text	Micro/Small mandatory, Medium recommended	All text mandatory
Hooping	Tight frame, center seam	Taut, never stretched	Stabilizer taut, garment relaxed
Sequencing	Center-out, bottom-up	Normal	Normal, low stitch count
Speed	600–650 small text	650–700 small text	600–650 everything
Needle point	Sharp 75/11	SES ballpoint 75/11 preferred	SES ballpoint 75/11 strongly recommended
Wide satin limit	Split > 8 mm	Split > 8 mm	Split > 7 mm; avoid very wide

8. TATAMI / FILL SOP — All Products

Addendum v1.0 — to be merged into Satin Stitch Embroidery SOP. Applies whenever the Satin SOP says "convert to fill," and to any solid shape or stroke wider than 10 mm.

8.1 Fill Size Bands (decided by ELEMENT SIZE, not letter height)

Band	Element Size	Typical Use
NARROW FILL	8–15 mm strokes	Converted wide-satin letters, thick borders
MEDIUM FILL	15–50 mm shapes	Standard logo blocks, large letters
LARGE FILL	> 50 mm areas	Backgrounds, big solid shapes

8.2 Settings by Product — TATAMI

Parameter	HAT (buckram)	POLO (pique)	BREATHABLE / STRETCH
Density (spacing, mm)	0.38–0.40	0.40–0.42	0.45–0.50 $\triangle$ start 0.45, untested
Stitch length (mm)	3.5–4.0	3.5–4.0	3.8–4.2 (fewer penetrations)
Underlay	Edge-run + zigzag, 2.5–3.0 mm spacing	Edge-run + double zigzag, 2.0–3.0 mm spacing	Edge-run + double zigzag 2.0 mm; add crosshatch for LARGE FILL
Underlay angle	90° to top stitch angle	90° to top stitch angle	90° to top stitch angle
Pull comp	0.15–0.20 mm	0.20 mm	0.20–0.30 mm (~10%) $\triangle$ start 0.20
Pattern	Standard tatami, offset/random pattern line	Offset/random pattern line	Offset/random pattern line — MANDATORY (aligned penetrations = visible tracks)
Machine speed	Normal	Normal	600–650 spm

$\triangle$ = not yet validated by a sample sew. Run the QC check in 8.5, then replace ranges with confirmed values and mark ✓ (same convention as the satin tables).

8.3 Universal Tatami Rules (all products)

1. **Vary stitch angles between adjacent large elements** (e.g., 45° / 90° / 135°). Identical angles make all pulls additive → directional rippling, worst on knits.
2. **Sequencing: center-out.** Sew large fills before outlines and details; path from the design center outward so displacement distributes evenly.
3. **Stacked stitching rule:** when text/script sews ON TOP of a fill, either (a) knock out the fill under the overlay, or (b) reduce fill density in the overlap region by ~20%. Never stack two full-density layers on knits. Topping over the fill is mandatory for MICRO/SMALL overlay text (see Section 5B).
4. **Fill-to-satin border:** sew fill first, border last; border overlaps fill edge 0.5–0.8 mm.
5. **Density is contrast-sensitive:** apply the Color-Contrast Density Rule (Section 4B) — high contrast → tighten 0.03–0.05 mm; tone-on-tone → values as-is.
6. **Stitch-count budget (art approval gate):** left-chest designs over ~25,000 stitches on BREATHABLE/STRETCH require art review — consider lighter fill types, larger open areas, or fewer stacked layers before production.

8.4 Product-Specific Notes

- **HAT:** large fills near the crown seam need a basting run first; keep fill angle off-vertical to avoid flagging on the curved frame.
- **POLO:** the double zigzag underlay is what allows the lighter top density — it builds a raft so the fill floats instead of gripping raw pique.
- **BREATHABLE / STRETCH:** two crossed layers of no-show mesh cutaway for any design over ~8,000 stitches (per Section 5 rule 1). Wash-test one sample before approving production — fill pucker on these knits often appears only after first wash.

8.5 Quality Check After Sample (fill objects)

- Fabric within 10 mm of the fill edge lies flat when garment is relaxed. Fail → reduce density by 0.03 mm OR check hooping stretch before changing digitizing.
- No visible fabric grin-through inside the fill at arm's length. Fail → tighten density 0.03 mm (check contrast rule first).
- No aligned penetration tracks visible under raking light. Fail → change pattern offset/randomization.
- Measure a filled letter's width vs artwork. Short by > 0.3 mm → increase pull comp +0.05 mm.
- POLO/STRETCH: wash-test before large orders.

8.6 Quick Reference — Satin vs Tatami Decision

Column / Stroke Width	Action
< 8 mm	Satin (per Satin SOP bands)
8–10 mm	Auto-split satin
> 10 mm	Tatami — always (this section)

Also recommended while merging: (1) standardize pull comp units in the satin tables — HAT/WIDE "115–118%" and POLO/VERY WIDE "115%" → convert to ~0.20 mm for consistency; (2) add example garments under each product heading, e.g., "Sport-Tek Micropique Sport-Wick → Section 4B"; (3) copy the Stacked Stitching Rule (8.3.3) into Section 6 Universal Rules since it applies to satin overlays too.